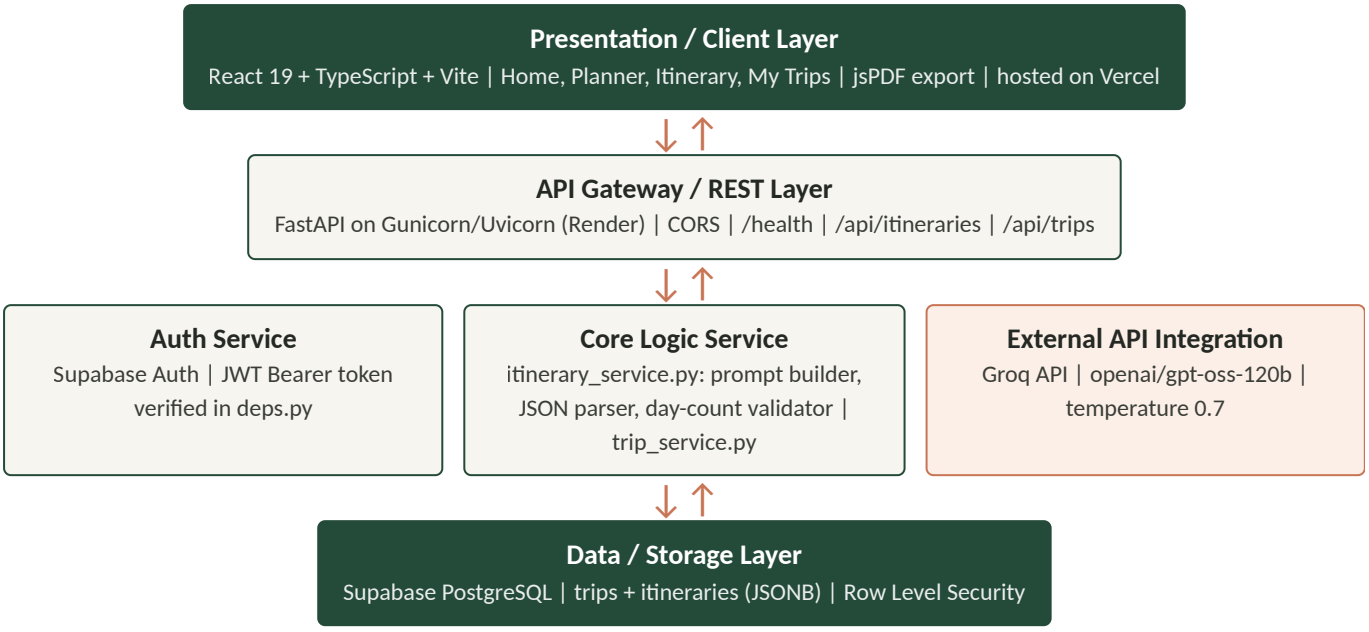
Solution Architecture

Date	13 November 2026
Team ID	0157CS241131
Project Name	AI Travel Itinerary Planner (Trip Genius AI)
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects preferences, shows loading/errors, renders day-wise plan, manages auth modal, saved trips and PDF export	React 19, TypeScript, Vite, jsPDF, supabase-js
API Gateway (REST layer)	Routes requests, validates payloads, enforces CORS, returns JSON and HTTP errors	FastAPI, Pydantic, Uvicorn, Gunicorn
Auth Service	Signs users in, issues JWT; backend verifies the token against Supabase	Supabase Auth, httpx
Core Logic Service	Builds system/user prompts, calls Groq, parses and validates JSON, handles timeouts/rate limits; trip CRUD	Python, groq SDK
Database	Persists trips and itineraries per user with RLS policies and indexes	Supabase PostgreSQL, JSONB

ER Diagram (Database)

auth.users
- **id** UUID (PK)

public.trips
- **id** UUID (PK)

public.itineraries
- **id** UUID (PK)

- email
- created_at

- user_id UUID (FK)
- title, destination
- start_date, end_date
- budget, travel_style
- preferences JSONB
- status (draft/planned/saved/completed)
- created_at, updated_at

- trip_id UUID (FK, UNIQUE)
- user_id UUID (FK)
- itinerary_data JSONB
- created_at, updated_at

Relationships: one user then many trips (1:N, ON DELETE CASCADE); one trip then one itinerary (1:1, UNIQUE trip_id, ON DELETE CASCADE). RLS: `auth.uid() = user_id` on all operations.